In the Java ecosystem, the two most-used tools for tracking, versioning, and deploying database changes are Flyway (https://flywaydb.org) and Liquibase (https://liquibase.org). Both of them are fully integrated with Spring Boot. This section will show you how to use Flyway.

5.4.1 Understanding Flyway: Version control for your database

Flyway is a tool that provides version control for your database. It offers a single source of truth for the version of your database's state and keeps track of any changes incrementally. It automates changes and lets you reproduce or roll back the state of a database. Flyway is highly reliable, safe to use in cluster environments, and supports several relational databases, including the cloud ones like Amazon RDS, Azure Database, and Google Cloud SQL.

NOTE In this section, I will introduce some features offered by Flyway, but I recommend you check the official documentation to discover all the powerful possibilities provided by this tool (https://flywaydb.org).

At its core, Flyway manages database changes. Any database change is called a *migration*, and migrations can be either *versioned* or *repeatable*. Versioned migrations are identified by a unique version number and are applied in order exactly once. For each *regular* versioned migration, you can also provide an optional *undo* migration to revert its effects (in case something goes wrong). They can be used to create, alter, or drop relational objects like schemas, tables, columns, and sequences or to correct data. On the other hand, repeatable migrations are applied every time their checksum changes. They can be used for creating or updating views, procedures, and packages.

Both types of migration can be defined in standard SQL scripts (useful for DDL changes) or Java classes (useful for DML changes, like data